

LAPORAN PROJECT DATA WAREHOUSE
UJIAN AKHIR SEMESTER 4



Disusun oleh:

Ahmad Hayyin Baihaki	(2341760199)
Azaria Cindy Sahasika	(2341760169)
Dimas Arya Sadewa	(2341760173)
Satriya Viar Citta Purnama	(2341760157)

SISTEM INFORMASI BISNIS
POLITEKNIK NEGERI MALANG
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1. Studi Kasus Pengelolaan Data

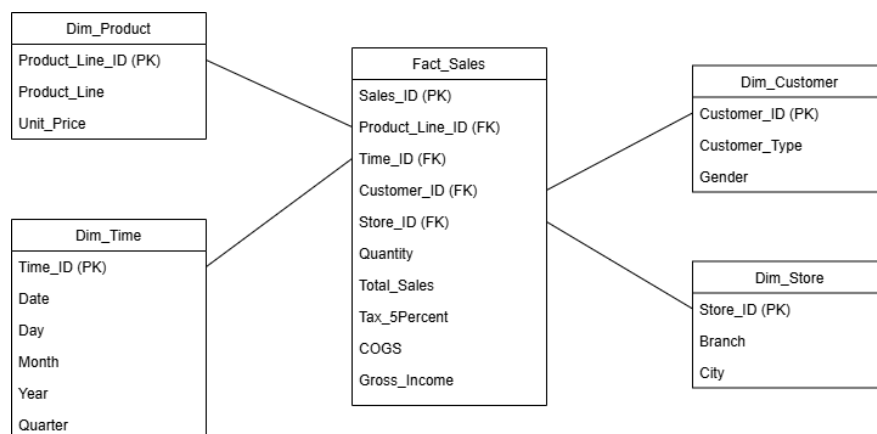
Studi kasus data yang kami gunakan adalah Data Penjualan dengan atribut invoice id, branch, city, customer type, gender, product line, unit price, quantity, tax 5%, total, date, cogs, dan gross income. Studi kasus kelompok kami yaitu menemukan produk yang paling laku terjual (berdasarkan jumlah atau total penjualan) pada bulan tertentu menggunakan dataset penjualan supermarket. Dataset ini memiliki informasi tentang:

- Product (product line)
- Waktu (date)
- Penjualan (quantity dan total)

2. Data Dummy Studi Kasus

Data dummy kami ambil dari Kaggle merupakan dataset tentang [penjualan supermarket](#). Dataset ini akan kami gunakan sebagai sumber data untuk proses ETL di Pentaho.

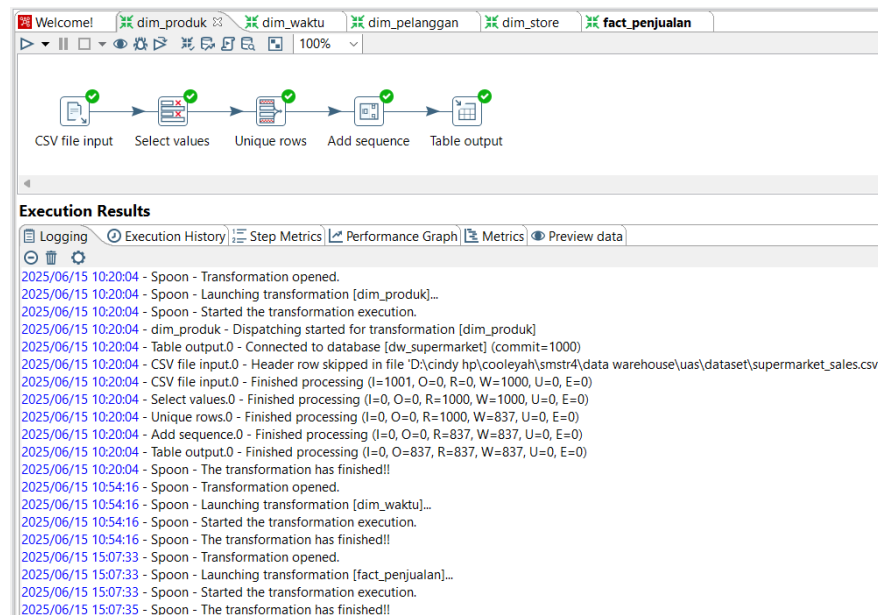
3. Rancangan Star Schema untuk Data Warehouse



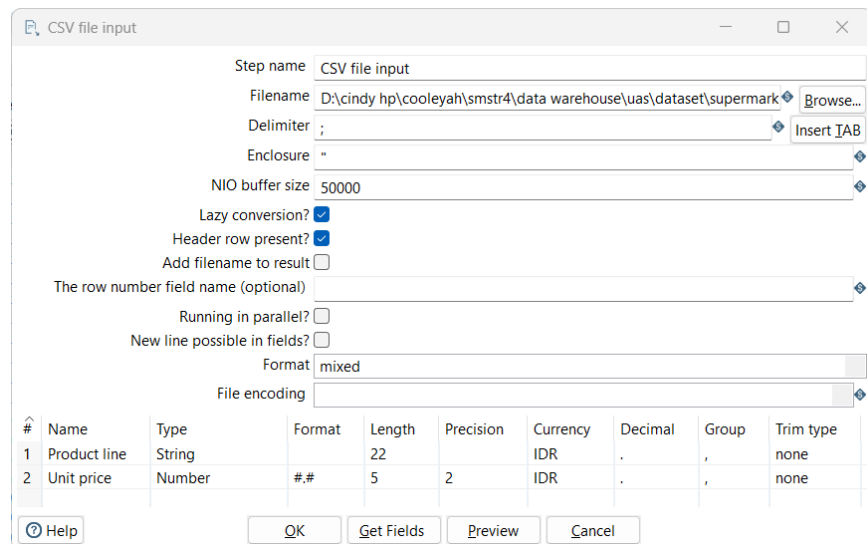
4. Rancangan Proyek ETL Berdasarkan Star Schema

a. Dim_Product

- Design konfigurasi transformasi di spoon pentaho



- Extract atau mengambil data dari data dummy supermarket_sales



- Transform atau melakukan pembersihan data

Select values

Step name: Select values

Select & Alter Remove Meta-data

Fields:

#	Fieldname	Rename to	Length	Precision
1	Product line			
2	Unit price			

Get fields to select

Edit Mapping

Include unspecified fields, ☐

Help OK Cancel

Unique rows

Step name: Unique rows

Settings

Add counter to ☐ Counter field

Redirect duplicate ☒ Error description

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	Product line	N

Help OK Cancel Get

Add sequence

Step name: Add sequence

Name of value: Product_Line_ID

Use a database to generate the sequence

Use DB to get sequence? ☐

Connection: conn_dw_supermaket Edit... New... Wizard...

Schema name Schemas...

Sequence name SEQ Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence? ☒

Counter name (optional): Product_Line_ID

Start at value: 1

Increment by: 1

Maximum value: 999999999

Help OK Cancel

- Load atau memasukkan data ke table dimensi produk

The screenshot shows the 'Table output' dialog box in SQL Developer. The 'Main options' tab is active, and the 'Database fields' section is selected. The 'Partition data over tables' checkbox is unchecked. The 'Use batch update for inserts' checkbox is checked. The 'Is the name of the table defined in a field?' checkbox is unchecked. The 'Return auto-generated key' checkbox is unchecked. The 'OK', 'Cancel', and 'SQL' buttons are at the bottom.

Table output

Step name: Table output

Connection: dw_supermarket [Edit...] [New...] [Wizard...]

Target schema: dw_supermarket [Browse...]

Target table: Dim_Product [Browse...]

Commit size: 1000

Truncate table: ☒

Ignore insert errors: ☐

Specify database fields: ☒

Main options | Database fields

Fields to insert:

#	Table field	Stream field
1	Product_Lin...	Product_Line...
2	Product_Line	Product line
3	Unit_Price	Unit price

[Get fields] [Enter field mapping]

[OK] [Cancel] [SQL]

[Help]

Product_Line_ID	Product_Line	Unit_Price
1	Health and beauty	74.69
2	Electronic accessories	15.28
3	Home and lifestyle	46.33
4	Health and beauty	58.22
5	Sports and travel	86.31
6	Electronic accessories	85.39
7	Home and lifestyle	73.56
8	Health and beauty	36.26
9	Food and beverages	54.84
10	Fashion accessories	14.48

b. Dim_Time

- Design konfigurasi tranformasi di spoon pentaho

The screenshot displays the Apache Pentaho Spoon interface. At the top, there are tabs for various data sources: dim_produk, dim_waktu, dim_pelanggan, dim_store, and fact_penjualan. Below the tabs, a workflow is configured with the following steps: CSV file input, Select values, Unique rows, Add sequence, Filter rows, Calculator, and Table output. Each step is represented by an icon and a label, with green checkmarks indicating successful execution.

Below the workflow, the 'Execution Results' section is visible, showing a log of the transformation process. The log includes the following entries:

```

2025/06/15 13:25:52 - Spoon - Transformation opened.
2025/06/15 13:25:52 - Spoon - Launching transformation [dim_waktu]...
2025/06/15 13:25:52 - Spoon - Started the transformation execution.
2025/06/15 13:25:52 - dim_waktu - Dispatching started for transformation [dim_waktu]
2025/06/15 13:25:52 - Table output.0 - Connected to database [conn_dw_supermarket] (commit=1000)
2025/06/15 13:25:53 - CSV file input.0 - Header row skipped in file 'D:\cindy hp\cooleyah\smstr4\data warehouse\uas\dataset\supermarket_sales.csv'
2025/06/15 13:25:53 - CSV file input.0 - Finished processing (I=1001, O=0, R=0, W=1000, U=0, E=0)
2025/06/15 13:25:53 - Select values.0 - Finished processing (I=0, O=0, R=1000, W=1000, U=0, E=0)
2025/06/15 13:25:53 - Unique rows.0 - Finished processing (I=0, O=0, R=1000, W=990, U=0, E=0)
2025/06/15 13:25:53 - Add sequence.0 - Finished processing (I=0, O=0, R=990, W=990, U=0, E=0)
2025/06/15 13:25:53 - Filter rows.0 - Finished processing (I=0, O=0, R=990, W=990, U=0, E=0)
2025/06/15 13:25:53 - Calculator.0 - Finished processing (I=0, O=0, R=990, W=990, U=0, E=0)
2025/06/15 13:25:53 - Table output.0 - Finished processing (I=0, O=0, R=990, W=990, U=0, E=0)
2025/06/15 13:25:53 - Spoon - The transformation has finished!!
2025/06/15 15:07:33 - Spoon - Transformation opened.
2025/06/15 15:07:33 - Spoon - Launching transformation [fact_penjualan]...
2025/06/15 15:07:33 - Spoon - Started the transformation execution.
2025/06/15 15:07:35 - Spoon - The transformation has finished!!
2025/06/15 15:19:26 - Spoon - Transformation opened.
2025/06/15 15:19:26 - Spoon - Launching transformation [dim_produk]...
2025/06/15 15:19:26 - Spoon - Started the transformation execution.
2025/06/15 15:19:26 - Spoon - The transformation has finished!!

```

- Extract atau mengambil data dari data dummy supermarket_sales

CSV file input

Step name: CSV file input

Filename: D:\cindy hp\cooleyah\smstr4\data warehouse\uas\dataset\supermark Browse...

Delimiter: ; Insert TAB

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	Date	Date	MM/dd/yyyy			IDR	.	,	none

Help OK Get Fields Preview Cancel

- Transform atau melakukan pembersihan data

Select values

Step name: Select values

Select & Alter Remove Meta-data

Fields:

#	Fieldname	Rename to	Length	Precision
1	Date			

Get fields to select Edit Mapping

Include unspecified fields, ordered ☐

Help OK Cancel

Unique rows

Step name: Unique rows

Settings

Add counter to output? ☐ Counter field

Redirect duplicate row ☒ Error description

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	Date	N

Help OK Cancel Get

Add sequence

Step name: Add sequence

Name of value: Time_ID

Use a database to generate the sequence

Use DB to get sequence? ☐

Connection: conn_dw_supermarket Edit... New... Wizard...

Schema name: Schemas...

Sequence name: SEQ_ Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence? ☒

Counter name (optional): Time_ID

Start at value: 1

Increment by: 1

Maximum value: 99999999

Help OK Cancel

Filter rows

Step name: Filter rows

Send 'true' data to step: Calculator

Send 'false' data to step:

The condition:

☐ +

Date IS NOT NULL

AND

(

Date >= [1900-01-01]

AND Date <= [2099-12-31]

)

Help OK Cancel

Calculator

Step name: Calculator

☒ Throw an error on non existing files

Fields:

#	New field	Calculation	Field A	Field B	Field C	Value type	Length	Precision	Remove	Conversion mask	Decimal symbol	Grouping symbol	Currency symbol
1	Day	Day of month of date A	Date			Date			N				
2	Month	Month of date A	Date			Integer			N				
3	Year	Year of date A	Date			Integer			N				

Help OK Cancel

- Load atau memasukkan data ke table dimensi waktu

Time_ID	Date	Day	Month	Year
1	2019-01-05	1970-01-01 07:00:00	1	2019
2	2019-03-08	1970-01-01 07:00:00	3	2019
3	2019-03-03	1970-01-01 07:00:00	3	2019
4	2019-01-27	1970-01-01 07:00:00	1	2019
5	2019-02-08	1970-01-01 07:00:00	2	2019
6	2019-03-25	1970-01-01 07:00:00	3	2019
7	2019-02-25	1970-01-01 07:00:00	2	2019
8	2019-02-24	1970-01-01 07:00:00	2	2019
9	2019-01-10	1970-01-01 07:00:00	1	2019
10	2019-02-20	1970-01-01 07:00:00	2	2019

c. Dim_customer

- Design konfigurasi tranformasi di spoon pentaho

The screenshot displays the Apache Pentaho Data Integration (Spoon) interface. At the top, a tabbed window shows several transformations: 'dim_produk', 'dim_waktu', 'dim_pelanggan', 'dim_store', and 'fact_penjualan'. The main workspace shows a sequence of five transformation steps: 'CSV file input', 'Select values', 'Unique rows', 'Add sequence', and 'Table output'. Below the workspace, the 'Execution Results' panel is active, showing a log of the transformation's execution. The log includes timestamps and status messages for each step, indicating that the transformation was successfully completed.

Execution Results

- 2025/06/15 10:54:16 - Spoon - Transformation opened.
- 2025/06/15 10:54:16 - Spoon - Launching transformation [dim_waktu]...
- 2025/06/15 10:54:16 - Spoon - Started the transformation execution.
- 2025/06/15 10:54:16 - Spoon - The transformation has finished!!
- 2025/06/15 11:26:59 - Spoon - Transformation opened.
- 2025/06/15 11:26:59 - Spoon - Launching transformation [dim_pelanggan]...
- 2025/06/15 11:26:59 - Spoon - Started the transformation execution.
- 2025/06/15 11:27:00 - dim_pelanggan - Dispatching started for transformation [dim_pelanggan]
- 2025/06/15 11:27:00 - Table output.0 - Connected to database [dw_supermarket] (commit=1000)
- 2025/06/15 11:27:00 - CSV file input.0 - Header row skipped in file 'D:\cindy hp\cooleyah\smstr4\data warehouse\uas\dataset\supermarket_sales.csv'
- 2025/06/15 11:27:00 - Select values.0 - Finished processing (I=0, O=0, R=1000, W=1000, U=0, E=0)
- 2025/06/15 11:27:00 - Unique rows.0 - Finished processing (I=0, O=0, R=1000, W=754, U=0, E=0)
- 2025/06/15 11:27:00 - Add sequence.0 - Finished processing (I=0, O=0, R=754, W=754, U=0, E=0)
- 2025/06/15 11:27:00 - Table output.0 - Finished processing (I=0, O=0, R=754, W=754, U=0, E=0)
- 2025/06/15 11:27:00 - Spoon - The transformation has finished!!
- 2025/06/15 15:19:26 - Spoon - Transformation opened.
- 2025/06/15 15:19:26 - Spoon - Launching transformation [dim_produk]...
- 2025/06/15 15:19:26 - Spoon - Started the transformation execution.
- 2025/06/15 15:19:26 - Spoon - The transformation has finished!!

- Extract atau mengambil data dari data dummy supermarket_sales

CSV file input

Step name: CSV file input

Filename: D:\cindy hp\cooleyah\smstr4\data warehouse\uas\dataset\supermark Browse...

Delimiter: ; Insert IAB

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	Customer type	String		6		IDR	.	,	none
2	Gender	String		6		IDR	.	,	none

Help
OK
Get Fields
Preview
Cancel

- Transform atau melakukan pembersihan data

Select values

Step name: Select values

Select & Alter Remove Meta-data

Fields:

#	Fieldname	Rename to	Length	Precision	
1	Customer type	Customer_Type			
2	Gender				

Get fields to select Edit Mapping

Include unspecified fields, ordered ☐

Help OK Cancel

Unique rows

Step name: Unique rows

Settings

Add counter to output? ☐ Counter field

Redirect duplicate row ☒ Error description

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	Customer_Type	N
2	Gender	N

Help OK Cancel Get

Add sequence

Step name: Add sequence

Name of value: Customer_ID

Use a database to generate the sequence

Use DB to get sequence? ☐

Connection: dw_supermarket Edit... New... Wizard...

Schema name Schemas...

Sequence name SEQ_ Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence? ☒

Counter name (optional): Customer_ID

Start at value: 1

Increment by: 1

Maximum value: 999999999

Help OK Cancel

- Load atau memasukkan data ke table dimensi pelanggan

Customer_ID	Customer_Type	Gender
1	Member	Female
2	Normal	Female
3	Normal	Male
4	Member	Male
5	Normal	Male
6	Member	Female
7	Normal	Female
8	Member	Female
9	Member	Male
10	Normal	Female

d. Dim_Store

- Design konfigurasi tranformasi di spoon pentaho

The screenshot displays the Apache Pentaho Data Integration (Spoon) interface. At the top, a tab bar shows several transformations: 'dim_produk', 'dim_waktu', 'dim_pelanggan', 'dim_store', and 'fact_penjualan'. The main workspace shows a workflow diagram with five steps: 'CSV file input', 'Select values', 'Unique rows', 'Add sequence', and 'Table output'. Each step is marked with a green checkmark, indicating successful execution.

Below the workflow diagram, the 'Execution Results' section is visible. It contains a log of events, including the opening and execution of transformations for 'dim_waktu', 'dim_pelanggan', and 'dim_store'. The log also shows the completion of the 'Table output' step, which is connected to a database named 'dw_supermarket'. The log entries provide timestamps and details about the execution process, such as the number of rows processed and the commit size.

- Extract atau mengambil data dari data dummy supermarket_sales

CSV file input

Step name: CSV file input

Filename: D:\cindy hp\cooleyah\smstr4\data warehouse\uas\dataset\supermark Browse...

Delimiter: ; Insert TAB

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	Branch	String		1		IDR	.	,	none
2	City	String		9		IDR	.	,	none

Help OK Get Fields Preview Cancel

- Transform atau melakukan pembersihan data

Select values

Step name: Select values

Select & Alter Remove Meta-data

Fields:

#	Fieldname	Rename to	Length	Precision
1	Branch			
2	City			

Get fields to select
Edit Mapping

Include unspecified fields, ordered ☐

Help OK Cancel

Unique rows

Step name: Unique rows

Settings

Add counter to output? ☐ Counter field

Redirect duplicate row ☒ Error description

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	Branch	N
2	City	N

Help OK Cancel Get

Add sequence

Step name: Add sequence

Name of value: Store_ID

Use a database to generate the sequence

Use DB to get sequence? ☐

Connection: dw_supermarket Edit... New... Wizard...

Schema name Schemas...

Sequence name SEQ_ Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence? ☒

Counter name (optional): Store_ID

Start at value: 1

Increment by: 1

Maximum value: 999999999

Help OK Cancel

- Load atau memasukkan data ke table dimensi toko

Table output

Step name: Table output

Connection: dw_supermarket [Edit...] [New...] [Wizard...]

Target schema: dw_supermarket [Browse...]

Target table: Dim_Store [Browse...]

Commit size: 1000

Truncate table: ☒

Ignore insert errors: ☐

Specify database fields: ☒

Main options

Database fields

Partition data over tables: ☐

Partitioning field: [dropdown]

Partition data per month: ☒

Partition data per day: ☐

Use batch update for inserts: ☒

Is the name of the table defined in a field?: ☐

Field that contains name of table: [dropdown]

Store the tablename field: ☒

Return auto-generated key: ☐

Name of auto-generated key field: [text field]

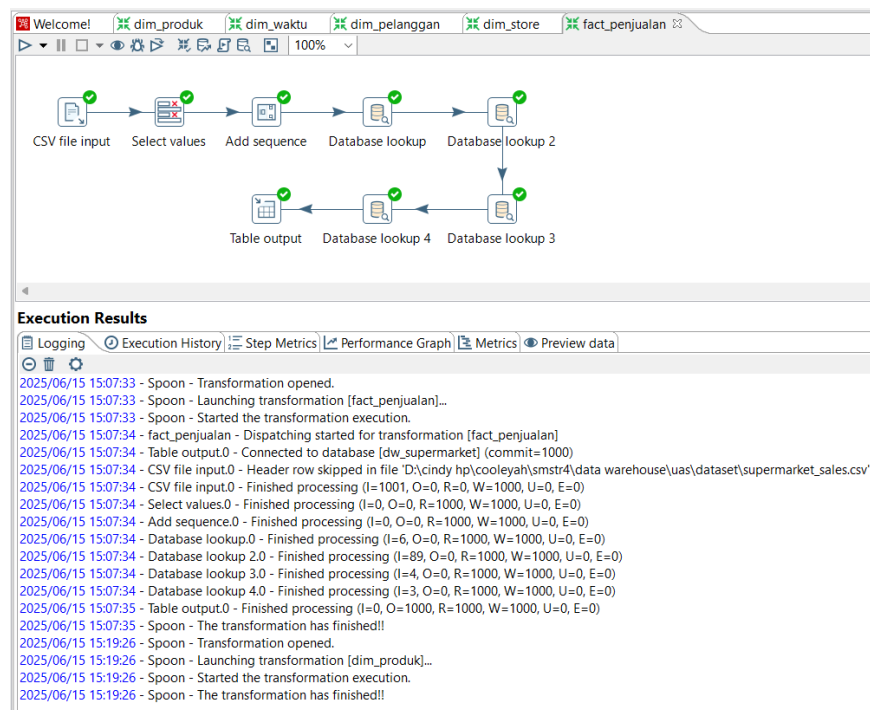
[Help] [OK] [Cancel] [SQL]

[illegible]

Store_ID	Branch	City
1	A	Yangon
2	C	Naypyitaw
3	A	Yangon
4	C	Naypyitaw
5	A	Yangon
6	C	Naypyitaw
7	A	Yangon
8	B	Mandalay
9	A	Yangon
10	B	Mandalay

e. Fact_Sales

- Design konfigurasi tranformasi di spoon pentaho



- Extract atau mengambil data dari data dummy supermarket_sales

CSV file input

Step name: CSV file input

Filename: D:\cindy hp\cooleyah\smstr4\data warehouse\uas\dataset\supermarket_sal Browse...

Delimiter: ; Insert TAB

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result ☐

The row number field name (optional)

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim type
1	Branch	String		1		IDR	.	,	none
2	City	String		9		IDR	.	,	none
3	Customer type	String		6		IDR	.	,	none
4	Gender	String		6		IDR	.	,	none
5	Product line	String		22		IDR	.	,	none
6	Unit price	Number	#,.	5	2	IDR	.	,	none
7	Quantity	Integer	#	15	0	IDR	.	,	none
8	Tax 5%	Number	#,.	7	4	IDR	.	,	none
9	Total	Number	#,##0.00	9	3	IDR	.	,	none
1..	Date	Date	MM/dd/yyyy			IDR	.	,	none
1..	cogs	Number	#,.	6	2	IDR	.	,	none
1..	gross income	Number	#,.	7	4	IDR	.	,	none

Help OK Get Fields Preview Cancel

- Transform atau melakuakn pembersihan data

Select values

Step name

Select values

Select & Alter

Remove

Meta-data

Fields :

#	Fieldname	Rename to	Length	Precision	
1	Branch				
2	City				
3	Customer type				
4	Gender				
5	Product line				
6	Unit price				
7	Quantity				
8	Tax 5%				
9	Total				
1..	Date				
1..	cogs				
1..	gross income				

Get fields to select

Edit Mapping

Include unspecified fields, ordered

☐

Help

OK

Cancel

Add sequence

Step name

Add sequence

Name of value

Sales_ID

Use a database to generate the sequence

Use DB to get sequence?

☐

Connection

dw_supermarket

Edit...

New...

Wizard...

Schema name

Schemas...

Sequence name

SEQ_

Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence?

☒

Counter name (optional)

Sales_ID

Start at value

1

Increment by

1

Maximum value

999999999

Help

OK

Cancel

Database lookup

Step name

Database lookup

Connection

dw_supermarket

Edit...

New...

Wizard...

Lookup schema

dw_supermarket

Browse...

Lookup table

dim_product

Browse...

Enable cache?

☒

Cache size in rows (0=cache)

0

Load all data from table

☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	Product_Line	=	Product line	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	Product_Line_ID			Integer

Do not pass the row if the lookup fails ☒

Fail on multiple results? ☐

Order by

Help

OK

Cancel

Get Fields

Get lookup fields

Database lookup

Step name

Database lookup 2

Connection

dw_supermarket

Edit...

New...

Wizard...

Lookup schema

dw_supermarket

Browse...

Lookup table

dim_time

Browse...

Enable cache?

☒

Cache size in rows (0=cache)

0

Load all data from table

☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	Date	=	Date	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	Time_ID			Integer

Do not pass the row if the lookup fails ☒

Fail on multiple results? ☐

Order by

Help

OK

Cancel

Get Fields

Get lookup fields

Database lookup

Step name: Database lookup 3

Connection: dw_supermarket Edit... New... Wizard...

Lookup schema: dw_supermarket Browse...

Lookup table: dim_customer Browse...

Enable cache? ☒

Cache size in rows (0=cache): 0

Load all data from table: ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	Customer_Type	=	Customer type	
2	Gender	=	Gender	

Values to return from the lookup table:

#	Field	New name	Default	Type
1	Customer_ID			Integer

Do not pass the row if the lookup fails ☒

Fail on multiple results? ☐

Order by:

Help OK Cancel Get Fields Get lookup fields

Database lookup

Step name: Database lookup 4

Connection: dw_supermarket Edit... New... Wizard...

Lookup schema: dw_supermarket Browse...

Lookup table: dim_store Browse...

Enable cache? ☒

Cache size in rows (0=cache): 0

Load all data from table: ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	Branch	=	Branch	
2	City	=	City	

Values to return from the lookup table:

#	Field	New name	Default	Type
1	Store_ID			Integer

Do not pass the row if the lookup fails ☒

Fail on multiple results? ☐

Order by:

Help OK Cancel Get Fields Get lookup fields

- Load atau memasukkan data ke table fakta penjualan

Table output

Step name: Table output

Connection: dw_supermarket Edit... New... Wizard...

Target schema: dw_supermarket Browse...

Target table: fact_sales Browse...

Commit size: 1000

Truncate table: ☒

Ignore insert errors: ☐

Specify database fields: ☒

Main options Database fields

Partition data over tables: ☐
 Partitioning field:

Partition data per month: ☒
 Partition data per day: ☐

Use batch update for inserts: ☒

Is the name of the table defined in a field?: ☐
 Field that contains name of table:
 Store the tablename field: ☒

Return auto-generated key: ☐
 Name of auto-generated key field:

Help OK Cancel SQL

Table output

Step name: Table output

Connection: dw_supermarket Edit... New... Wizard...

Target schema: dw_supermarket Browse...

Target table: fact_sales Browse...

Commit size: 1000

Truncate table: ☒

Ignore insert errors: ☐

Specify database fields: ☒

Main options Database fields

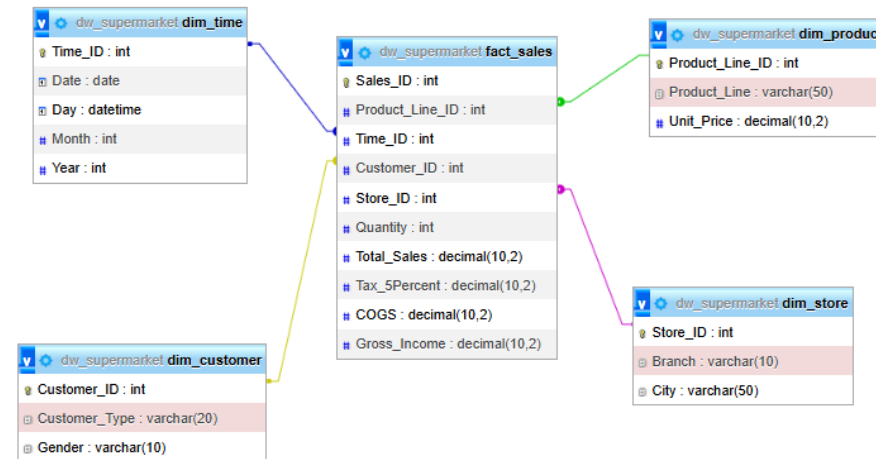
Fields to insert:

#	Table field	Stream field
1	Sales_ID	Sales_ID
2	Product_Lin...	Product_Line...
3	Time_ID	Time_ID
4	Customer_ID	Customer_ID
5	Store_ID	Store_ID
6	Quantity	Quantity
7	Total_Sales	Total
8	Tax_SPercent	Tax 5%
9	COGS	cogs
1..	Gross_Inco...	gross income

Get fields

Enter field mapping

Help OK Cancel SQL



Sales_ID	Product_Line_ID	Time_ID	Customer_ID	Store_ID	Quantity	Total_Sales	Tax_5Percent	COGS	Gross_Income
1	1	1	1	1	7	5489715.00	261.42	522.83	261.42
2	2	2	2	2	5	8022.00	3.82	76.40	3.82
3	3	3	3	1	7	3405255.00	162.16	324.31	162.16
4	1	4	4	1	8	489048.00	23.29	465.76	23.29
5	5	5	3	1	7	6343785.00	302.09	604.17	302.09
6	2	6	3	2	7	6276165.00	298.87	597.73	298.87
7	2	7	1	1	6	433692.00	20.65	413.04	20.65
8	3	8	2	2	10	77238.00	36.78	735.60	36.78
9	1	9	1	1	2	76146.00	3.63	72.52	3.63
10	9	10	1	8	3	172746.00	8.23	164.52	8.23

5. Analisis menggunakan query SQL

```

SELECT
    dp.Product_Line,
    SUM(fs.Quantity) AS Total_Quantity,
    SUM(fs.Total_Sales) AS Total_Sales
FROM
    fact_sales fs
JOIN
    dim_time dt ON fs.Time_ID = dt.Time_ID
JOIN
    dim_product dp ON fs.Product_Line_ID = dp.Product_Line_ID
WHERE
    dt.Month = 3
GROUP BY
    dp.Product_Line
ORDER BY
    Total_Quantity DESC, Total_Sales DESC
LIMIT 1;
  
```


Product_Line	Total_Quantity	Total_Sales
Home and lifestyle	364	46227699.00

Pada bulan maret atau bulan 3 prodduk yang palin laku terjual adalah ‘Home and lifestyle’ dengan total jumlah penjualan 364 items. Kita dapat melihat produk terlaris lainnya dengan mengubal ‘dt.Month = 3’ dengan bulan lainnya.